Targeted Dephasing as a Causal Handle on Local Occupation Loss in an Open Bose–Hubbard Chain

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We show that sites of high local occupation variance $F_i = \langle \hat{n}_i^2 \rangle - \langle \hat{n}_i \rangle^2$ function as causal handles on future local occupation loss in an open one-dimensional Bose–Hubbard chain under Lindblad dephasing. Applying additional dephasing at the top-k highest-F_i sites produces more local occupation loss than an equal-budget random targeting at $J/U \in \{0.30, 0.40\}$, with 95% bootstrap confidence intervals excluding zero across all three tested time horizons ($\tau \in \{1, 2, 3\}$) and both tested chain lengths ($L \in \{6, 7\}$). The effect is absent or reversed at the lowest tested coupling ($J/U = 0.12$) and size-sensitive at $J/U = 0.20$; the handle is a regime-specific property of the dephasing operator, not a generic one. The site-resolved spatial response is more consistent with nonlocal redistribution of occupation change than with a local-instability mechanism. All results use exact Lindblad time evolution in the fixed-particle-number sector; no approximations are employed.

I. INTRODUCTION

The one-dimensional Bose–Hubbard model is a standard setting for studying the interplay of quantum tunnelling and on-site interactions in interacting boson systems [1, 2]. At zero temperature the model exhibits a quantum phase transition between a superfluid phase (large J/U) and a Mott-insulating phase (small J/U), controlled by the ratio of tunnelling amplitude J to on-site interaction U [3, 4]. The clean-system phase diagram and its experimental realisation in optical lattices [5] have been extensively studied [4].

In the presence of dissipation—modelled by local dephasing in the Lindblad–Gorini–Kossakowski–Sudarshan (LGKS) framework [6–8]—the system is driven away from the ground state and develops spatially heterogeneous fluctuation structure. Open quantum systems of this type arise naturally in cold-atom experiments where coupling to photon-field fluctuations causes local dephasing of on-site number states [9, 10]. More broadly, the engineering of dissipation as a resource for state preparation and quantum control has attracted significant theoretical and experimental attention [11–14].

In classical lattice systems, local structural variables can serve as causal handles: targeted perturbations at structurally selected sites produce measurably different future outcomes from matched-budget perturbations at random sites [15]. Whether analogous causal handles exist in open quantum systems—where dynamics are governed by a Lindblad master equation and the state is a density matrix rather than a classical configuration—is an empirical question that has received little direct attention.

This paper tests one specific instance. Under uniform dephasing, sites in the Bose–Hubbard chain develop different local occupation variances $F_i = \langle \hat{n}_i^2 \rangle - \langle \hat{n}_i \rangle^2$, reflecting the degree of local number fluctuation. We ask whether selecting the highest-F_i sites for additional dephasing produces a different pattern of future occupation loss than applying the same total dephasing budget to randomly chosen sites. The question is about the causal accessibility of future dynamical outcomes through structurally targeted interventions, not about the equilibrium phase diagram.

II. MODEL AND METHODS

A. Bose–Hubbard Hamiltonian

The Hamiltonian for a chain of L sites with open boundary conditions is

$$\hat{H} = -J \sum_{i=1}^{L-1} \left(\hat{a}_i^\dagger \hat{a}_{i+1} + \text{h.c.} \right) + \frac{U}{2} \sum_{i=1}^{L} \hat{n}_i (\hat{n}_i - 1), \quad (1)$$

where J is the tunnelling amplitude, U the on-site interaction, $\hat{n}_i = \hat{a}_i^\dagger \hat{a}_i$, and the local Fock space is truncated at $n_{\max} = 3$ occupations per site. The system is studied at half-filling ($N = \lfloor L/2 \rfloor$ particles) in the fixed-particle-number sector. Open boundary conditions are used throughout, so that the local variance field F_i can develop nontrivial spatial structure across the chain.

B. Hilbert space

The computation is restricted to the sector of exactly N particles distributed across L sites, each with at most $n_{\max} = 3$ occupations. The dimension of this sector is $\binom{L+N-1}{N}$ reduced by the $n_{\max}$ constraint; for $L = 6$, $N = 3$ the dimension is $D = 56$, and for $L = 7$, $N = 3$ it is $D = 84$. All operators are constructed as dense matrices in this basis.

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C. Lindblad dynamics

The density matrix $\hat{\rho}$ evolves under the Lindblad master equation

$$\frac{d\hat{\rho}}{dt} = -i[\hat{H}, \hat{\rho}] + \sum_{i=1}^{L} \gamma_i \Big(\hat{L}_i \hat{\rho} \hat{L}_i^\dagger - \tfrac{1}{2}\{\hat{L}_i^\dagger \hat{L}_i,\, \hat{\rho}\} \Big), \qquad (2)$$

where $\hat{L}_i = \hat{n}_i$ is a dephasing jump operator and γ_i is the dephasing rate at site i. Baseline dephasing is uniform: $\gamma_i = \gamma = 0.1$ for all sites. Since $\hat{L}_i = \hat{n}_i$ commutes with the total number operator $\hat{N} = \sum_i \hat{n}_i$, the dissipator decoheres number-state superpositions without changing any local expectation value $\langle \hat{n}_i \rangle$; both local and total particle number are conserved at all times.

The Lindblad equation is converted to a superoperator acting on the vectorised density matrix in the row-major convention. Time evolution is computed by the action of the matrix exponential on the state vector, using `scipy.sparse.linalg.expm_multiply` [16], which avoids forming the full propagator. No Trotter decomposition, stochastic unravelling, or other approximation is used.

D. Burn-in state

The initial state is the ground state of $\hat{H}$, obtained by exact diagonalisation. This pure state is then evolved under uniform baseline dephasing for a burn-in period $t_{\text{burn}} = 5.0$ (in units of $\hbar/U$). The resulting mixed state $\hat{\rho}_{\text{burn}}$ serves as the starting point for all interventions within a given condition. The burn-in ensures that the state has developed spatially heterogeneous fluctuation structure under dissipation before any intervention is applied.

E. Handle definition

The handle variable is the local occupation variance

$$F_i = \langle \hat{n}_i^2 \rangle - \langle \hat{n}_i \rangle^2, \qquad (3)$$

evaluated on the post-burn-in state $\hat{\rho}_{\text{burn}}$. This quantity identifies sites with larger number fluctuations. Under open boundary conditions, F_i is typically largest at the central sites of the chain, where the tunnelling network is most connected, and smallest at the edges. The top-k sites by F_i ($k = \lceil L/3 \rceil$) are selected for intervention. The selected set is fixed by the post-burn-in state and is therefore deterministic within each condition.

F. Intervention protocol

The **targeted intervention** applies additional dephasing $\gamma_{\text{extra}} = 0.5$ at the selected high-F_i sites, keeping all other sites at the baseline rate γ. The **random control** applies the same total extra budget ($k \times \gamma_{\text{extra}}$) to k randomly chosen sites. In both cases the total dissipative budget is identical. Because $\hat{L}_i = \hat{n}_i$ commutes with the total number operator, $\langle \hat{n}_i \rangle$ is unchanged at the moment of intervention by construction; the occupation profile is held fixed at the start of each arm.

The targeted intervention is deterministic: for a given condition, the post-burn-in state and the selected sites are the same, so the targeted evolution is computed once per (condition, τ) pair. Only the random arm varies across trials.

G. Target

The primary target is local future occupation loss at the intervention sites:

$$\mathcal{L}_{\text{local}}(\tau) = \sum_{i \in \mathcal{S}} \max\big(0,\ \langle \hat{n}_i \rangle_t - \langle \hat{n}_i \rangle_{t+\tau}\big), \qquad (4)$$

where $\mathcal{S}$ is the set of selected sites and τ is the measurement horizon. The $\max(\cdot, 0)$ clip counts only sites that *lose* occupation; gains at other sites do not offset local losses.

H. Design and statistical analysis

Four tunnelling-to-interaction ratios $J/U \in \{0.12, 0.20, 0.30, 0.40\}$ are tested, spanning from deep Mott-like to intermediate superfluid-like regimes. Three horizons $\tau \in \{1, 2, 3\}$ and two chain lengths $L \in \{6, 7\}$ are used. For each condition, $N_{\text{trials}} = 100$ independent random-site draws are compared against the single targeted intervention. The test statistic is the mean difference in $\mathcal{L}_{\text{local}}$ between targeted and random interventions. A 95% bootstrap confidence interval (1000 resamples, trial-level resampling) is constructed for each condition. The decision rule is that a condition passes if the lower bound of the 95% CI exceeds zero.

III. RESULTS

A. Main result at $L = 6$

Table I reports the local-loss difference for all conditions at $L = 6$. At $L = 6$, the targeted intervention produces more local occupation loss than the random control throughout $J/U \in \{0.20, 0.30, 0.40\}$, with all confidence intervals in that range excluding zero, while the broader cross-size regime map (Sec. III C) shows that $J/U = 0.20$ is transitional. The lowest tested coupling $J/U = 0.12$ is uniformly negative: the targeted intervention produces *less* local loss than matched-budget random targeting at all three horizons.

TABLE I. Local-loss difference (targeted − random) at $L = 6$, $N_{\text{trials}} = 100$ per condition. Format: mean [95% CI]. $J/U = 0.12$ is uniformly negative (boundary regime).

J/U	$\tau = 1$	$\tau = 2$	$\tau = 3$
0.12	−0.002 [−0.002, −0.001]	−0.002 [−0.003, −0.002]	−0.002 [−0.003, −0.001]
0.20	0.001 [0.000, 0.001]	0.006 [0.005, 0.007]	0.011 [0.009, 0.012]
0.30	0.013 [0.012, 0.014]	0.029 [0.027, 0.031]	0.040 [0.036, 0.043]
0.40	0.019 [0.017, 0.021]	0.040 [0.036, 0.043]	0.047 [0.042, 0.051]

The effect increases with J/U from the negative boundary at 0.12 through a transitional regime at 0.20 and into the robust positive pocket at 0.30–0.40. The magnitude of the positive difference also increases with the time horizon τ, consistent with a cumulative dynamical effect rather than a transient perturbation.

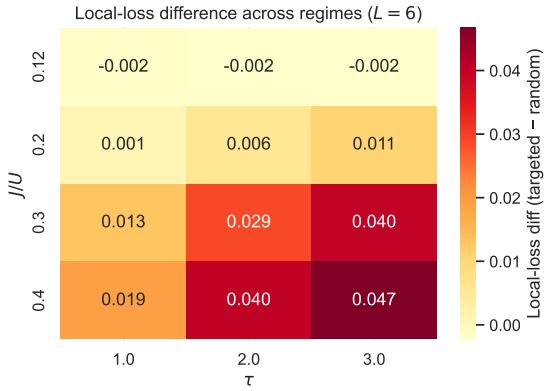

FIG. 1. Local-loss difference across J/U and τ at $L = 6$. The handle effect is positive in the intermediate and stronger coupling regime and absent or reversed at the lowest tested coupling.

B. Robustness across chain length

Table II compares pooled results at $J/U \in \{0.30, 0.40\}$ for $L = 6$ and $L = 7$. Both couplings pass at both chain lengths with confidence intervals excluding zero, confirming that the positive pocket is not a single-size artefact.

The intermediate coupling $J/U = 0.20$ is size-sensitive: it passes at $L = 6$ across all horizons but is not robust at $L = 7$, where only the longest horizon ($\tau = 3$) produces a positive CI. This places $J/U = 0.20$ in a transitional regime whose handle effectiveness depends on system size.

C. Regime map

The results define three regimes within the tested parameter range:

TABLE II. Pooled local-loss difference for $L = 6$ and $L = 7$ at $J/U = 0.30$ and 0.40. Format: mean [95% CI], pooling trial-level differences across $\tau \in \{1, 2, 3\}$.

Condition	$L = 6$	$L = 7$
$J/U = 0.30$	0.027 [0.025, 0.029]	0.030 [0.027, 0.032]
$J/U = 0.40$	0.035 [0.032, 0.038]	0.047 [0.044, 0.051]

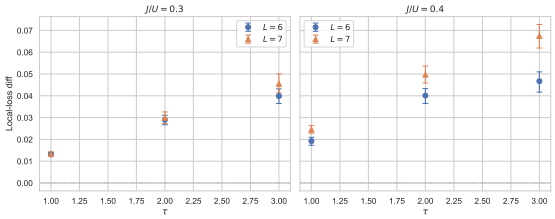

FIG. 2. Local-loss difference at $J/U = 0.30$ and 0.40 for $L = 6$ and $L = 7$. Error bars are 95% bootstrap CIs. The positive pocket persists across both chain lengths.

Negative regime ($J/U = 0.12$): The targeted intervention produces less local loss than random targeting at both chain lengths and all horizons. In this deep Mott-like regime the occupation-variance handle does not identify causally effective sites.

Transitional regime ($J/U = 0.20$): Positive at $L = 6$ but not robust at $L = 7$. The handle begins to function but its effectiveness is size-dependent.

Positive pocket ($J/U \in \{0.30, 0.40\}$): The handle is consistently positive across both chain lengths and all horizons. This is the regime where the occupation-variance selector reliably identifies sites whose perturbation produces excess local loss.

The existence of a negative boundary regime and a transitional zone is itself informative: the handle is not a generic property of the dephasing operator but depends on the dynamical regime of the system.

D. Spatial response pattern

Fig. 3 shows the site-resolved occupation change under targeted versus random intervention at $J/U = 0.30$, $L = 6$, $\tau = 2$. The bars show the mean site-resolved occupation change under the deterministic targeted intervention and averaged across the 100 matched-budget random interventions.

The observed pattern is consistent with a nonlocal, redistributive response. Under targeted intervention, the selected sites (shaded) tend to show *less* negative occupation change than under random intervention: selected-site loss is reduced relative to random targeting. At the same time, non-selected sites show increased occupation loss under targeted intervention. The net effect is that the intervention changes the spatial pattern of occupation change across the chain rather than simply amplifying loss at the targeted sites themselves.

This is the opposite of what a purely local-instability mechanism would predict, in which extra dephasing at a high-F_i site would increase loss specifically at that site. The data are more consistent with a picture in which perturbing high-F_i sites alters the subsequent spatial distribution of occupation change across the chain.

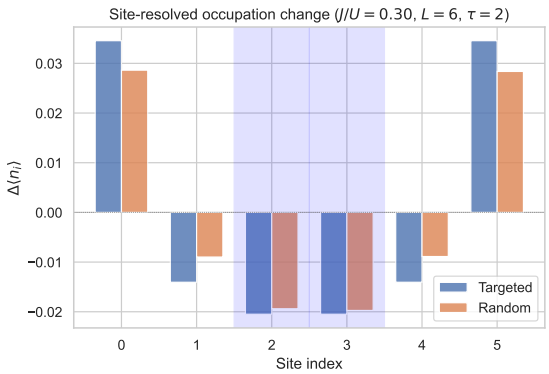

FIG. 3. Mean site-resolved occupation change under the targeted intervention and across matched-budget random interventions ($J/U = 0.30$, $L = 6$, $\tau = 2$). Negative values correspond to occupation loss. Shaded bands mark selected sites. The spatial response is consistent with nonlocal redistribution.

IV. DISCUSSION

a. Summary. In an open Bose–Hubbard chain, sites of high local occupation variance serve as causal handles in the intermediate and stronger coupling regime: targeting them with additional dephasing produces more local future occupation loss than matched-budget random targeting at $J/U \in \{0.30, 0.40\}$, robustly across both tested chain lengths and all tested horizons. The effect is absent or reversed at the lowest tested coupling ($J/U = 0.12$) and transitional at $J/U = 0.20$.

b. Mechanism. The site-resolved response is more consistent with nonlocal redistribution than with local instability. Perturbing high-F_i sites is associated with a redistribution of subsequent occupation change across the chain. A complete mechanistic account is beyond the scope of the present analysis.

c. What is not claimed. We do not claim that high-F_i targeting is optimal among all possible intervention strategies, that the effect extends to all coupling regimes (it does not hold at $J/U = 0.12$ and is only transitional at $J/U = 0.20$), that the mechanism generalises to other quantum systems or larger chains, or that the occupation-variance handle corresponds to any equilibrium observable.

d. Scope and limitations. The causal-handle effect is regime- and size-dependent in the present design, with the clearest fully robust support in the $J/U = 0.30$–0.40 regime. The Hilbert-space dimension of the fixed-N sector grows combinatorially with L and $n_{\max}$, limiting exact Lindblad evolution to small chains. The tested range ($L \leq 7$, $n_{\max} = 3$) establishes the effect in the positive pocket but does not access the thermodynamic limit. Whether the positive pocket persists, broadens, or shifts at larger system sizes is an open question requiring approximate methods such as matrix-product-operator Lindblad evolution [14].

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DATA AND CODE AVAILABILITY

All simulation code and raw data are available at github.com/kunalb541/BH.

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